A small-parameter limit for higher-dimensional Liouville first passage percolation

Abstract

Let $\lambda_d(\zeta)$ be the distance exponent for Liouville first passage percolation in $\mathbb{R}^d$, with logarithmic covariance normalized to have coefficient one. For each fixed $d \geq 2$, we prove that $\zeta^{-4/3}\lambda_d(\zeta)$ has a finite limit as $\zeta \downarrow 0$. We identify the limit as the subadditive rate of a Gaussian variational problem on paths with $d-1$ transverse coordinates. Together with the previously established planar lower bound, the limit is strictly positive. The upper comparison controls all paths by consecutive partitions and a Gaussian estimate uniform in the field cutoff. For the lower comparison, equal-length polygonal replacements preserve an invariant class of expected weighted occupation measures, allowing the Gaussian gain to be iterated through arbitrarily many scales.

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1 Introduction

Liouville first passage percolation assigns a random length to a path by exponentiating a regularized log-correlated Gaussian field. Its distance exponent describes the power of the regularization scale needed to normalize the resulting distances. In this note we prove that, in every fixed dimension $d \geq 2$, the exponent has an asymptotic expansion $\lambda_d(\zeta) = \kappa_d \zeta^{4/3} + o(\zeta^{4/3})$ as $\zeta \downarrow 0$. The constant is characterized by a Gaussian optimization problem, and the previously established planar lower bound implies $\kappa_d > 0$.

1.1 Definitions and main result

Fix an integer $d \geq 2$. Let $\kappa : \mathbb{R}^d \to [0, \infty)$ be a smooth, compactly supported radial function with $\|\kappa\|_{L^2} = 1$. For $t > 0$, set $K_t(z) = t^{-(d+1)/2}\kappa(z/t)$, and let W be white noise on $\mathbb{R}^d \times (0, \infty)$ with Lebesgue control measure. We work with the regularization

$$H_{a,b}(z) = \int_a^b \int_{\mathbb{R}^d} K_t(z - y)\, W(dy, dt), \qquad 0 < a < b < \infty\,. \tag{1.1}$$

In particular, $\operatorname{Var} H_{\varepsilon,1}(z) = \log \varepsilon^{-1}$. Put $U = (-2, 2)^d$ and $e_1 = (1, 0, \ldots, 0)$. For $\zeta > 0$, define

$$D^{\zeta}_{\varepsilon,d} = \inf_{P:0\to e_1,\ P\subset U} \int_P e^{\zeta H_{\varepsilon,1}(z)}\, |dz|\,. \tag{1.2}$$

The infimum is over rectifiable paths, and integrals count arclength with multiplicity. We use the fixed-parameter exponent $\lambda_d(\zeta)$, characterized for all sufficiently small $\zeta > 0$ by

$$\frac{\log D^{\zeta}_{\varepsilon,d}}{\log \varepsilon} \longrightarrow \lambda_d(\zeta) \quad \text{in probability as } \varepsilon \downarrow 0\,. \tag{1.3}$$

The normalization, the fixed-parameter input, and the relation with other regularizations are discussed in Section A.

Let $V_{n,d}$ be the space of continuous functions $f : [0, 1] \to \mathbb{R}^{d-1}$ that vanish at both endpoints and are affine on every dyadic interval of length 2^{-n}. Set

$$\mathcal{E}(f) = \frac{1}{2} \int_0^1 |f'(x)|^2\, dx\,. \tag{1.4}$$

Let $\mathcal{G}_d$ be the centered Gaussian process, with $\mathcal{G}_d(0) = 0$, whose covariance is

$$\operatorname{Cov}(\mathcal{G}_d(f), \mathcal{G}_d(g)) = \pi \int_0^1 \left(|f(x)| + |g(x)| - |f(x) - g(x)|\right) dx\,. \tag{1.5}$$

Its construction is given in Section 2. Define

$$A_{n,d} = \mathbb{E} \sup_{f\in V_{n,d}} \{\mathcal{G}_d(f) - \mathcal{E}(f)\}\,. \tag{1.6}$$

Theorem 1.1. *For each fixed integer $d \geq 2$, the sequence $(A_{n,d})_{n\geq 1}$ is finite and subadditive. Moreover,*

$$\lim_{\zeta\downarrow 0} \frac{\lambda_d(\zeta)}{\zeta^{4/3}} = \kappa_d, \qquad \kappa_d := \inf_{n\geq 1} \frac{A_{n,d}}{n \log 2} = \lim_{n\to\infty} \frac{A_{n,d}}{n \log 2} < \infty\,. \tag{1.7}$$

The previously established planar lower bound $\lambda_2(\zeta) \geq c\zeta^{4/3}$ implies $\kappa_d > 0$ for every $d \geq 2$.

The planar lower bound used in Theorem 1.1 is the prior input from [5, Theorem 1.1]. We use it only to prove positivity; the existence and variational identification of the limit follow from the comparisons proved here. Lemma A.1 identifies the circle-average convention of that input with (1.2). The fixed-parameter existence of λ_d is supplied by [2, Proposition 1.1]; the small-parameter question appears as Problem 7.4 of that paper.

The characterization also gives a comparison between dimensions:

$$(d-1)^{2/3}\kappa_2 \le \kappa_d \le (d-1)\kappa_2\,. \tag{1.8}$$

This follows by comparing the vector Gaussian process with independent scalar copies; see Lemma 2.3. All assertions concern a fixed dimension. No uniform error as $d \to \infty$ is asserted.

Throughout, the dimension d is fixed. A constant C_d may depend on d and the fixed kernel, but not on the block size; a constant $C_{n,d}$ may also depend on n. Both may change from line to line. We write ν_Q for normalized arclength on a curve of positive length, and $\mathcal{L}$ for Lebesgue probability measure on $[0,1]$. The change of variables

$$\delta = \zeta^{2/3}, \qquad \zeta = \delta^{3/2}, \qquad \zeta^{4/3} = \delta^2 \tag{1.9}$$

will be used in both comparisons. Every small δ limit is taken with all block parameters fixed.

1.2 Proof outline

The scale $\delta = \zeta^{2/3}$ can be seen by perturbing a unit segment into the graph of δf. Its excess Euclidean length is $\delta^2\mathcal{E}(f) + o(\delta^2)$. The difference between the field averages along the graph and the segment has variance of order δ, with limiting covariance (1.5). The corresponding random term in the log cost has size $\zeta\sqrt{\delta}$. Balancing this with δ^2 gives $\delta = \zeta^{2/3}$ and a gain of order $\zeta^{4/3}$. Lemma 3.2 proves the covariance statement, including its normalization in every dimension.

The main issue is to compare this local optimization with paths spanning arbitrarily many scales, while retaining its leading coefficient. Subadditivity is available for the Gaussian quantities $A_{n,d}$: interpolate on a coarse mesh, subtract the interpolation, and compare with independent Gaussian processes for the two parts. This yields the rate κ_d. We prove that the normalized LFPP exponent is squeezed between two approximations to this same rate.

Upper comparison. A consecutive first-exit partition gives a finite tree of subpaths. The log cost is bounded below by a weighted sum of the log length excesses and the changes in field averages when chords are replaced by their child polygons. At a node with small excess, the child polygon is a forward graph, and its penalized mean converges to the variational optimization. We compare truncated fields with the all-scale field before taking this limit, so the estimate is uniform over arbitrarily fine cutoffs.

To make the estimate simultaneous for all paths, we record the endpoint locations and excess bins along each chain. The entropy of these records and the covariance between scales contribute an error $O_d(n^{3/4})$ for a block of n dyadic scales. A node with larger excess pays for its more weakly controlled field gain through its deterministic length cost. The resulting bound is (4.1); after dividing by the block length and taking $n \to \infty$, its right side tends to κ_d.

Lower comparison. Choose a finite family of almost optimal transverse graphs with a prescribed energy bound. A small oscillatory perturbation makes all replacement edges have exactly the same length. This permits exact alignment of independent scale bands under every subsequent edge rescaling.

The selected polygon is determined by the current band, but the final path will be further refined. We control that change through a deterministic expected weighted occupation measure. Its horizontal marginal stays within $O(\delta^2)$ of uniform measure in W_∞, even though its transverse support has width $O(\delta)$. A prefix-average estimate shows that exponential weighting preserves this horizontal control. The same class is therefore available at every depth, and the current band's Gaussian saving survives all later refinements. Conditional expectation gives an exact recursion for expected path costs, without requiring independence between different edges. This proves (5.1).

In both comparisons, the block size is fixed first, the field cutoff tends to zero next, and then $\zeta \downarrow 0$. Only after these limits do we let the block size tend to infinity. Thus no interchange of the cutoff and small-parameter limits is needed.

Proof of Theorem 1.1. Lemma 2.1 proves finiteness, subadditivity, and the existence of κ_d. The upper and lower comparisons give (4.25) and (5.29), respectively. These inequalities identify the limit with κ_d.

For positivity, apply the proved identification in dimension two and the prior lower bound $\lambda_2(\zeta) \geq c\zeta^{4/3}$ to obtain $\kappa_2 \geq c$. Lemma 2.3 then gives $\kappa_d \geq (d-1)^{2/3}\kappa_2 > 0$. □

Section 2 develops the Gaussian variational problem. Section 3 proves the transport, graph-limit, and scale-band estimates. Sections 4 and 5 give the two LFPP comparisons. Appendix A checks the fixed-parameter input and regularization conventions.

2 The Gaussian variational problem

The transverse displacement of a nearly straight path has a quadratic Euclidean cost. Its leading random contribution has increments proportional to the square root of the displacement. The balance between these two terms gives the power 4/3. We first make this variational problem precise and show that its mean has a limiting rate per logarithmic scale.

Write $m = d-1$ and let σ_m be uniform probability measure on S^{m-1}. Set $b_m = \int_{S^{m-1}} |\theta_1|\, \sigma_m(d\theta) > 0$. For $a \in \mathbb{R}$, define $J_a(t) = \mathbf{1}_{(0,a]}(t)$ if $a \geq 0$ and $J_a(t) = -\mathbf{1}_{(a,0]}(t)$ if $a < 0$. If $\mathcal{W}$ is white noise on $[0,1] \times S^{m-1} \times \mathbb{R}$ with control measure $dx\, \sigma_m(d\theta)\, dt$, then

$$\mathcal{G}_d(f) = \sqrt{\frac{2\pi}{b_m}}\, \mathcal{W}((x,\theta,t) \mapsto J_{\theta\cdot f(x)}(t)) \tag{2.1}$$

has covariance (1.5). Indeed, $\int_{\mathbb{R}} J_a J_b = (|a| + |b| - |a-b|)/2$ and rotational invariance gives $\int |\theta \cdot v|\, d\sigma_m = b_m |v|$. In particular,

$$\mathbb{E}[(\mathcal{G}_d(f) - \mathcal{G}_d(g))^2] = 2\pi \int_0^1 |f-g|\, dx\,. \tag{2.2}$$

We always use separable versions, continuous on bounded subsets of $V_{n,d}$.

We will use three Gaussian comparison facts. The Sudakov–Fernique inequality also holds with the same deterministic mean at each index: if $\mathbb{E}(X_s - X_t)^2 \leq \mathbb{E}(Y_s - Y_t)^2$, then

$$\mathbb{E} \sup_t (X_t + a_t) \leq \mathbb{E} \sup_t (Y_t + a_t)\,. \tag{2.3}$$

For a supremum T of Gaussian linear functionals with squared representing norms at most v, Gaussian concentration gives

$$\log \mathbb{E} e^{s(T - \mathbb{E}T)} \leq \frac{s^2 v}{2}, \qquad s \geq 0\,. \tag{2.4}$$

Deterministic means do not affect this bound. We also use Dudley's entropy integral for the expected supremum. These facts can first be applied to finite sets and then extended by separable approximation; see [1, 4, 3].

Lemma 2.1. *The numbers $A_{n,d}$ are finite. For $a, b > 0$ and $n, r \geq 1$,*

$$\mathbb{E} \sup_{f \in V_{n,d}} \{a\mathcal{G}_d(f) - b\mathcal{E}(f)\} = a^{4/3} b^{-1/3} A_{n,d}, \tag{2.5}$$

$$A_{n+r,d} \leq A_{n,d} + A_{r,d}\,. \tag{2.6}$$

Consequently, the last limit in (1.7) exists and belongs to $[0, \infty)$.

Proof. On the energy ball $\{\mathcal{E}(f) \leq R\}$, the Poincaré inequality and (2.2) give $\sup_f \operatorname{Var} \mathcal{G}_d(f) \leq C_d\sqrt{R}$. The same metric is bounded by $C_d \|f - g\|_{H_0^1}^{1/2}$. Finite-dimensional covering and Dudley's inequality therefore give a finite expected supremum on every energy ball. Since $(\mathcal{G}_d(cf))_f \overset{d}{=} (\sqrt{c}\,\mathcal{G}_d(f))_f$ for $c > 0$, the expected supremum on the radius R energy ball is $R^{1/4}$ times its value at $R = 1$. On the shells $k \leq \mathcal{E}(f) < k + 1$, concentration bounds the probability that this supremum exceeds a fixed positive multiple of k by $C_{n,d} \exp(-c_{n,d} k^{3/2})$ for sufficiently large k. Summation over the shells proves integrability after subtracting $\mathcal{E}(f)$. Substituting $f = (a/b)^{2/3} g$ proves (2.5).

For $f \in V_{n+r,d}$, let $g \in V_{n,d}$ interpolate f at the coarse mesh points and put $v = f - g$. The derivative of g is constant on each coarse interval, whereas v vanishes at its endpoints. Hence $\mathcal{E}(f) = \mathcal{E}(g) + \mathcal{E}(v)$. For two such decompositions, the triangle inequality gives

$$2\pi \int |(g + v) - (\widetilde{g} + \widetilde{v})| \leq 2\pi \int |g - \widetilde{g}| + 2\pi \int |v - \widetilde{v}|\,.$$

Compare $\mathcal{G}_d(g + v)$ with $\mathcal{G}_d^{(1)}(g) + \mathcal{G}_d^{(2)}(v)$ for independent copies of the process, using (2.3) with mean $-\mathcal{E}(g) - \mathcal{E}(v)$. The comparison supremum separates. Its coarse part has mean $A_{n,d}$. On a coarse interval of length $\ell = 2^{-n}$, the substitution $v(a + \ell x) = \ell u(x)$ multiplies both the Gaussian process and the energy by ℓ. The processes on disjoint intervals are independent by (2.1); thus the fine part has mean $2^n \ell A_{r,d} = A_{r,d}$. This proves (2.6). Fekete's lemma gives the limit, since $0 \leq A_{n,d} \leq n A_{1,d}$. □

The lower comparison will choose among finitely many paths with a common upper bound on their energy. We therefore replace the energy penalty by a prescribed energy ball. Define

$$w_{n,d} = \mathbb{E} \sup_{\substack{f \in V_{n,d} \\ \mathcal{E}(f) \leq 1}} \mathcal{G}_d(f), \qquad e_{n,d} = (w_{n,d}/4)^{4/3}, \qquad \beta_{n,d} = 3e_{n,d}\,. \tag{2.7}$$

Lemma 2.2. *There is a constant $C_d < \infty$ such that*

$$0 \leq A_{n,d} - \beta_{n,d} \leq C_d\sqrt{n}, \qquad e_{n,d} \leq C_d n\,. \tag{2.8}$$

For every sufficiently large n, there is a finite set $\mathcal{F}_{n,d} \subset V_{n,d}$ such that

$$\max_{f \in \mathcal{F}_{n,d}} \mathcal{E}(f) \leq e_{n,d}, \quad \mathbb{E} \max_{f \in \mathcal{F}_{n,d}} \mathcal{G}_d(f) \geq 4e_{n,d} - 1, \quad \log(2|\mathcal{F}_{n,d}|) \leq C_d n 2^n\,. \tag{2.9}$$

In particular, $\beta_{n,d}/(n \log 2) \to \kappa_d$.

Proof. Put $M_n(r) = \sup_{\mathcal{E}(f)\le r} \mathcal{G}_d(f)$. Homogeneity gives $\mathbb{E}M_n(r) = r^{1/4} w_{n,d}$. Maximizing $w_{n,d} r^{1/4} - r$ yields the value $3e_{n,d}$ at $r = e_{n,d}$, so $\beta_{n,d} \le A_{n,d}$. Also $\sup_{\mathcal{E}(f)\le r} \operatorname{Var} \mathcal{G}_d(f) \le C_d \sqrt{r}$. For $0 < \tau < 1/2$, divide the energy into intervals $[k, k+1]$ and use (2.4) with $s = 1$ to obtain

$$\begin{aligned} A_{n,d} &\le (1-\tau)^{-1/3} \beta_{n,d} + 1 + \mathbb{E} \sup_{k\ge 0} \{ M_n(k+1) - \mathbb{E}M_n(k+1) - \tau k \} \\ &\le (1-\tau)^{-1/3} \beta_{n,d} + 1 + \log \sum_{k\ge 0} \exp\{ C_d \sqrt{k+1} - \tau k \} \\ &\le (1-\tau)^{-1/3} \beta_{n,d} + C_d/\tau \,. \end{aligned}$$

In the last line, use $C_d\sqrt{k+1} \le \tau k/2 + C_d'/\tau$ and sum the geometric series. Since $\beta_{n,d} \le A_{n,d} \le nA_{1,d}$, choosing τ of order $n^{-1/2}$ proves (2.8).

To obtain the finite set, cover the energy ball $\{\mathcal{E}(f) \le e_{n,d}\}$ in the H_0^1 norm by balls of radius $\eta_n = c_d 2^{-4n} (n+2)^{-4}$, with centers in the energy ball. Its dimension is $(d-1)(2^n - 1)$ and its radius is at most $C_d\sqrt{n}$. The number J of centers satisfies

$$J \le \left(1 + C_d \sqrt{n}/\eta_n\right)^{(d-1)(2^n-1)} .$$

The canonical metric bound $C_d \|f-g\|_{H_0^1}^{1/2}$ and the entropy integral, applied inside these balls and then to their union, bound the expected error of the maximum by

$$C_d \sqrt{2^n \eta_n \log(2 + C_d\sqrt{n}/\eta_n)} \,.$$

This is at most one for an appropriate $c_d > 0$. Since $\mathbb{E}M_n(e_{n,d}) = 4e_{n,d}$, the centers give (2.9). □

Lemma 2.3. *For every $d \ge 2$ and $n \ge 1$,*

$$(d-1)^{2/3} A_{n,2} \le A_{n,d} \le (d-1) A_{n,2} \,. \tag{2.10}$$

The same inequalities hold for the limiting rates κ_d.

Proof. Write $f = (f_1, \dots, f_m)$ with $m = d-1$, and let $\mathcal{G}_2^{(1)}, \dots, \mathcal{G}_2^{(m)}$ be independent copies of the scalar process. The inequalities

$$m^{-1/2} \sum_{j=1}^m |f_j - g_j| \le |f - g| \le \sum_{j=1}^m |f_j - g_j|$$

and (2.3) compare $\mathcal{G}_d(f)$ with $m^{-1/4} \sum_j \mathcal{G}_2^{(j)}(f_j)$ from below and $\sum_j \mathcal{G}_2^{(j)}(f_j)$ from above. The energy is $\sum_j \mathcal{E}(f_j)$, so both comparison suprema separate by coordinate. Formula (2.5) gives $m\, m^{-1/3} A_{n,2}$ for the lower comparison and $mA_{n,2}$ for the upper one. Divide by $n \log 2$ and take the limit. □

3 Estimates for the Gaussian field

The comparison with the variational problem uses averages of the field along nearby curves. A bound in terms of the distance between the curves controls both the local approximation and the covariance between different scales. We prove this transport bound first, then identify the transverse limit and isolate the estimates for independent scale bands.

All constants in this section may depend on d and the fixed kernel κ. This dependence will be suppressed in the notation C_d. They do not depend on a block size n unless this is indicated explicitly.

3.1 Logarithmic covariance and transport

Put $C = \kappa * \kappa$, so C is smooth, radial, compactly supported, nonnegative, and $C(0) = 1$. Directly from (1.1),

$$\mathrm{Cov}(H_{a,b}(z), H_{a,b}(w)) = \int_a^b C((z-w)/t)\frac{dt}{t}\,. \tag{3.1}$$

Disjoint scale intervals give independent fields. If $T(z) = x + rOz$, where O is orthogonal and $r > 0$, then $H_{ra,rb} \circ T$ has the same law as $H_{a,b}$. These statements include joint laws at all spatial points.

For a compactly supported signed measure η of total mass zero and finite absolute logarithmic energy, define $\langle h, \eta\rangle$ by the white-noise integral over all scales. Its covariance is

$$\mathrm{Cov}(\langle h, \eta\rangle, \langle h, \widetilde{\eta}\rangle) = \iint \log|z-w|^{-1}\,\eta(dz)\,\widetilde{\eta}(dw)\,. \tag{3.2}$$

To check the coefficient, radiality and a change of variable give, for $r, s > 0$,

$$\int_0^\infty (C(re_1/t) - C(se_1/t))\frac{dt}{t} = \log(s/r)\,.$$

For example, truncate the integral at a, b, change variables separately in the two terms, and let $a \downarrow 0$, $b \uparrow \infty$, using $C(0) = 1$. The mass-zero property cancels the additive constant in the covariance. Smooth approximation then proves (3.2) for the stated measures. In particular, the covariance on mass-zero measures is exactly invariant under similarities.

The white-noise representation also gives the cutoff domination

$$\mathrm{Var}\langle H_{a,b}, \eta\rangle \le \mathrm{Var}\langle h, \eta\rangle\,. \tag{3.3}$$

We will apply this to differences of indices as well. It permits Gaussian comparison with h for every cutoff, without taking a limit in the cutoff inside an expected supremum.

A probability measure μ has a linear growth bound at scale R with constant B if

$$\mu(B(z,t)) \le B\min(1, t/R), \qquad z \in \mathbb{R}^d, \quad t > 0\,. \tag{3.4}$$

Here and below, balls in growth estimates are Euclidean balls. Normalized arclength on a segment of length comparable to R has this property. For two probability measures, a displacement coupling of size u means a coupling supported on pairs at Euclidean distance at most u.

Lemma 3.1. *Let $R \ge r > 0$, and let $\mu_R, \nu_R, \mu_r, \nu_r$ be compactly supported probability measures. Suppose μ_R, ν_R satisfy (3.4) at scale R, and μ_r, ν_r satisfy it at scale r, all with constant B. Suppose the two pairs have displacement couplings of sizes u and v, respectively. Then, for every scale interval $(a, b) \subset (0, \infty)$, including the all-scale field,*

$$\left|\mathrm{Cov}(\langle H_{a,b}, \mu_R - \nu_R\rangle, \langle H_{a,b}, \mu_r - \nu_r\rangle)\right| \le C_{d,B}\frac{\sqrt{uv}}{R}\,. \tag{3.5}$$

In particular, $\mathrm{Var}\langle H_{a,b}, \mu_R - \nu_R\rangle \le C_{d,B}u/R$.

Proof. Write $\eta_R = \mu_R - \nu_R$ and $\eta_r = \mu_r - \nu_r$. Compact support of the kernel, the growth bound, and a displacement coupling give

$$\|\eta_R * K_t\|_\infty \le \frac{C_{d,B}}{R}\,t^{-(d-1)/2}\min(1, u/t), \tag{3.6}$$

$$\|\eta_r * K_t\|_1 \le C_d t^{(d-1)/2}\min(1, v/t)\,. \tag{3.7}$$

For (3.6) without the factor u/t, integrate $|K_t|$ against each measure and use (3.4). If $u \le t$, write the difference of translated kernels as the integral of a derivative along the coupling segment. The derivative is bounded by $C_d t^{-(d+3)/2}$ and supported within distance $C_d t$ of the reference point; (3.4) supplies the extra factor t/R. For (3.7), use $\|K_t\|_1 = C_d t^{(d-1)/2}$ and $\|\nabla K_t\|_1 = C_d t^{(d-3)/2}$.

The covariance is the integral of $\langle \eta_R * K_t, \eta_r * K_t \rangle_{L^2}$ over the scale interval. Its absolute value is bounded by

$$\frac{C_{d,B}}{R} \int_0^\infty \min(1, u/t) \min(1, v/t)\, dt = \frac{C_{d,B}}{R}\, u_* (2 + \log(v_*/u_*)) \le \frac{2C_{d,B}}{R} \sqrt{uv}\,,$$

where $u_* = \min(u, v)$ and $v_* = \max(u, v)$. The zero-displacement cases follow by continuity. The growth bound implies finite logarithmic energy, so smooth approximation justifies the same calculation for curve measures. □

3.2 The transverse graph limit

For a continuous function $f : [0, 1] \to \mathbb{R}^{d-1}$, let

$$\mu_{\delta f} = \int_0^1 \delta_{(x, \delta f(x))}\, dx\,.$$

This is horizontal Lebesgue measure lifted to the graph; it is not normalized arclength. Its horizontal projection gives the growth bound (3.4) at scale one with a universal constant, regardless of the slope of f.

Lemma 3.2. *For bounded continuous $f, g : [0, 1] \to \mathbb{R}^{d-1}$,*

$$\lim_{\delta \downarrow 0} \delta^{-1} \operatorname{Var} \langle h, \mu_{\delta f} - \mu_{\delta g} \rangle = 2\pi \int_0^1 |f - g|\, dx\,. \tag{3.8}$$

On every compact subset of $V_{n,d}$, the processes $-\delta^{-1/2} \langle h, \mu_{\delta f} - \mu_0 \rangle$ converge in law, in the topology of continuous functions, to $\mathcal{G}_d(f)$. Expected suprema converge as well.

Proof. Expand the variance using the four logarithmic kernels in (3.2) and make the change of variable $y = x + \delta u$. After division by δ, the integrand is

$$\frac{1}{2} \log \frac{(u^2 + |f(x) - g(x + \delta u)|^2)(u^2 + |g(x) - f(x + \delta u)|^2)}{(u^2 + |f(x) - f(x + \delta u)|^2)(u^2 + |g(x) - g(x + \delta u)|^2)}\,,$$

integrated over $0 < x < 1$ and $-x/\delta < u < (1 - x)/\delta$. For almost every x and u, it converges to $\log(1 + |f(x) - g(x)|^2/u^2)$. If $\|f\|_\infty + \|g\|_\infty \le M$, its absolute value is bounded by $2 \log(1 + 4M^2/u^2)$, an integrable function of u. Dominated convergence and the identity

$$\int_{\mathbb{R}} \log(1 + a^2/u^2)\, du = 2\pi a, \qquad a \ge 0\,,$$

prove (3.8). The identity follows by differentiating in $a > 0$ and then using the value at $a = 0$.

Polarization proves convergence of the finite-dimensional distributions. Coupling graph points with the same horizontal coordinate and applying Lemma 3.1 gives

$$\delta^{-1} \operatorname{Var} \langle h, \mu_{\delta f} - \mu_{\delta g} \rangle \le C_d \|f - g\|_\infty\,. \tag{3.9}$$

On a compact finite-dimensional family, polynomial covering numbers in the uniform norm therefore give tightness by the Gaussian entropy integral. The same estimate and Gaussian concentration give uniform integrability of the suprema. This proves both remaining assertions. □

The next lemma makes the arclength and moving-knot replacements precise. The parameter space is fixed and finite dimensional; a bound on knot displacement alone would not ensure continuity of a process indexed by an arbitrary choice of knots. We write W_∞ for the infimum, over couplings, of the essential supremum of the Euclidean displacement.

Lemma 3.3. *Fix n, put $K = 2^n$, and let F be a nonempty compact subset of $V_{n,d}$. Fix $0 \le S < \infty$ and let $\mathcal{B}_S$ consist of vectors $b = (b_0, \dots, b_K) \in \mathbb{R}^{K+1}$ with $b_0 = b_K = 0$ and $\max_i |b_i| \le S$. For sufficiently small δ, let $Q_\delta(f, b)$ be the polygon with ordered vertices*

$$\Big(i/K + \delta^2 b_i,\, \delta f(i/K)\Big), \qquad 0 \le i \le K\,.$$

Then

$$\mathbb{E} \sup_{\substack{f \in F \\ b \in \mathcal{B}_S}} \Big| \delta^{-1/2} \langle h, \nu_{Q_\delta(f,b)} - \mu_{\delta f} \rangle \Big| \longrightarrow 0\,. \tag{3.10}$$

Consequently, the processes $-\delta^{-1/2}\langle h, \nu_{Q_\delta(f,b)} - \mu_0\rangle$, indexed by $F \times \mathcal{B}_S$, have the same continuous-process limit and expected-supremum limit as the horizontal graph processes of Lemma 3.2, with the latter viewed as constant in b. The same conclusion holds after restricting to continuous parameter choices $b = b_\delta(f) \in \mathcal{B}_S$; no convergence of b_δ is required.

Proof. All constants in this proof may depend on n, d, F, S. Write $|f-g|_* = \max_i |f(i/K) - g(i/K)|$ and $|b - c|_* = \max_i |b_i - c_i|$. The longitudinal edge lengths belong to $[1/(2K), 2/K]$ for small δ, and the transverse slopes are $O(\delta)$. Thus all the normalized arclength measures have a common linear growth bound.

The length of edge i is

$$\Big[(K^{-1} + \delta^2(b_i - b_{i-1}))^2 + \delta^2 |f(i/K) - f((i-1)/K)|^2\Big]^{1/2}\,.$$

Differentiating this expression on a fixed bounded set of vertex parameters shows that its variation is at most $C\delta^2(|f-g|_* + |b-c|_*)$. The same estimate holds for the normalized times at which arclength parametrizations visit vertices. Matching these times and interpolating between consecutive vertices therefore gives

$$W_\infty\big(\nu_{Q_\delta(f,b)}, \nu_{Q_\delta(g,c)}\big) \le C(\delta|f-g|_* + \delta^2|b-c|_*)\,. \tag{3.11}$$

Here the additional $O(\delta^2|f-g|_*)$ term from the vertex times is absorbed in the first term. This argument compares the normalized arclength maps at the same parameter, which is a displacement coupling.

For a fixed (f, b), the normalized arclength times differ from i/K by $O(\delta^2)$, as do the horizontal knots. Matching the same parameter with $x \mapsto (x, \delta f(x))$ gives

$$W_\infty\big(\nu_{Q_\delta(f,b)}, \mu_{\delta f}\big) \le C\delta^2\,.$$

Let $D_\delta(f, b) = \delta^{-1/2}\langle h, \nu_{Q_\delta(f,b)} - \mu_{\delta f}\rangle$. Transport, (3.9), and (3.11) imply

$$\sup_{f,b} \operatorname{Var} D_\delta(f, b) \le C\delta,$$

$$\operatorname{Var}(D_\delta(f, b) - D_\delta(g, c)) \le C\big(|f-g|_* + \delta|b-c|_*\big)\,.$$

The second bound gives polynomial covering numbers, with degree and constants independent of δ, on the fixed compact parameter set. The canonical diameter is $O(\sqrt{\delta})$ by the first bound. Applying the Gaussian entropy integral to the family and its negatives now proves (3.10); Gaussian concentration also gives uniform integrability of its suprema. The process is continuous in the vertex parameters. Combining this uniform error with Lemma 3.2 proves the assertions about process convergence and continuous restrictions. □

3.3 Independent scale bands

The lower comparison chooses a replacement path using one scale band and then refines it using finer, independent bands. The next estimates quantify the errors from discarding the very fine and very coarse scales. The last estimate also applies to occupation measures with atoms, which will be important under iteration.

Lemma 3.4. *The following bounds hold for compactly supported measures, with constants depending only on d, κ, and the indicated growth constant.*

(a) If μ, ν satisfy (3.4) at scale one, then

$$\operatorname{Var}\langle H_{0,a}, \mu - \nu\rangle \le C_d a\,. \tag{3.12}$$

(b) If $f, g \in H_0^1((0,1);\mathbb{R}^{d-1})$ and $\rho = \delta L$, then

$$\delta^{-1}\operatorname{Var}\langle H_{\rho,\infty}, \mu_{\delta f} - \mu_{\delta g}\rangle \le \frac{C_d}{L}\|f-g\|_{L^2}^2\,. \tag{3.13}$$

The expected supremum in absolute value of $\delta^{-1/2}\langle H_{\rho,\infty}, \mu_{\delta f} - \mu_0\rangle$, over $\mathcal{E}(f) \le E_0$, is at most $C_d\sqrt{E_0/L}$.

(c) If ν satisfies (3.4) at scale one and μ has a displacement coupling of size $u \le a$ with ν, then

$$\operatorname{Var}\langle H_{a,b}, \mu - \nu\rangle \le C_d u^2/a\,. \tag{3.14}$$

No growth bound on μ below scale a is required.

(d) Suppose $\nu_1, \dots, \nu_J$ each have a displacement coupling of size a with a measure satisfying (3.4) at scale one. For $H = H_{a,b}$ with $b \le 1$,

$$\mathbb{E}\sup_{\substack{1\le j\le J\\ x\in\mathbb{R}}}\left|\int_{\{z_1\le x\}} H(z)\,\nu_j(dz)\right| \le C_d\sqrt{b}\bigl(1+\sqrt{\log(2J)}\bigr)\,. \tag{3.15}$$

Proof. For (a), the squared L^2 norm of $\mu * K_t$ equals $\iint t^{-1}C((z-w)/t)\,\mu(dz)\mu(dw)$. The growth bound makes this at most C_d for $t \le 1$, and the same bound remains valid for $t > 1$. Integrate over $0 < t < a$ and apply $\|p-q\|_2^2 \le 2\|p\|_2^2 + 2\|q\|_2^2$.

For (b), apply the fundamental theorem of calculus in the transverse coordinates. The absolute covariance of two directional transverse derivatives of the scale t kernel is bounded by

$$C_d t^{-3}\mathbf{1}_{\{|x-x'|\le C_d t\}}\,.$$

Indeed, differentiation of $t^{-1}C((z-w)/t)$ gives the factor t^{-3}, and a nonzero term requires the horizontal coordinates to differ by at most $C_d t$. Young's inequality in the horizontal coordinate gives

$$\|(\mu_{\delta f} - \mu_{\delta g}) * K_t\|_2^2 \le C_d\delta^2 t^{-2}\|f-g\|_2^2\,.$$

Integrating over $t \ge \rho$ proves (3.13). For the supremum, compare by (2.3) with $C_d L^{-1/2}$ times $d-1$ independent horizontal white noises paired with f. Their expected squared dual H_0^1 norm is finite, since $\sum_{k\ge1}(\pi k)^{-2} < \infty$. Cauchy–Schwarz then bounds the expected supremum by $C_d\sqrt{E_0/L}$. Applying this argument to each sign bounds the absolute supremum.

For (c), use the displacement coupling and integrate the kernel derivative. For $t \geq a \geq u$, the reference measure ν supplies the linear growth bound in the derivative's support. This gives

$$\|(\mu - \nu) * K_t\|_\infty \leq C_d u\, t^{-(d+1)/2}, \qquad \|(\mu - \nu) * K_t\|_1 \leq C_d u\, t^{(d-3)/2}\,.$$

The product is $C_d u^2 t^{-2}$. Integrating over $a < t < b$ proves (3.14).

For (d), the coupling gives $\nu_j(B(z,r)) \leq C_d r$ for $r \geq a$. Since the covariance kernel C_H of H is nonnegative, (3.1) implies

$$\sup_z \int C_H(z,w)\, \nu_j(dw) \leq C_d b\,.$$

Consequently, for every positive measure $\eta \leq \nu_j$,

$$\mathrm{Var}\langle H, \eta\rangle \leq C_d b\, \eta(\mathbb{R}^d)\,. \tag{3.16}$$

The difference between two nested horizontal prefixes is such a measure. Indexing prefixes by their ν_j-mass bounds the canonical distance by $C_d\sqrt{b|s-t|}$. This remains valid when the horizontal marginal has atoms: the attained masses form a subset of $[0,1]$. A maximal separated set therefore has at most $1 + C_d b/r^2$ elements at canonical scale r. Include the J families and their negatives in the covering bound and apply the entropy integral. This proves (3.15). □

4 The upper comparison

To bound the exponent from above, we need a lower bound for the cost of every path, including paths chosen after observing the entire field. We partition each path into consecutive pieces and compare the field average along the resulting polygons with that along their chords. Polygons with small length excess are governed by the Gaussian variational problem. The remaining polygons have an excess that pays for their possible random gain.

We will prove that, for all sufficiently large n,

$$\limsup_{\zeta \downarrow 0} \frac{\lambda_d(\zeta)}{\zeta^{4/3}} \leq \frac{\alpha_n A_{n,d} + C_d n^{3/4}}{(n-1)\log 2}, \qquad \alpha_n = (1 + n^{-1/4})^{2/3}(1 - n^{-1/4})^{-1/3}\,. \tag{4.1}$$

Fix $K = 2^n$, $\varepsilon = 2^{-N}$ with $N \geq 1$, and abbreviate $H_N = H_{\varepsilon,1}$. In this section $n \geq 3$.

4.1 Consecutive path partitions

At a node of the partition tree, let P_u be a subpath with endpoints x_u, y_u and chord length $R_u > 0$. If $R_u > \varepsilon$, cut at successive first times when displacement from the current starting point equals $r = R_u/K$. If a nonconstant residual piece has chord length at most $r/2$, merge it with the preceding completed piece; otherwise retain it. A constant residual can be discarded. There is always a completed piece, since the endpoint displacement of P_u is $R_u > r$.

Every child v satisfies

$$\frac{R_u}{2K} \leq R_v \leq \frac{3R_u}{2K}, \qquad P_v \subset \overline{B}(x_v, 4R_v)\,. \tag{4.2}$$

Indeed, an unmerged first-exit piece lies in the radius r ball about its start. A retained residual has chord length greater than $r/2$ and lies in the radius r ball about its start. A merged piece lies in the radius $2r$ ball about its start and its chord has length at least $r/2$. All but possibly the last

child have chord length R_u/K. At the root, use the containment radius $\max(4, 2\sqrt{d})$, since $P \subset U$ and $R_{\text{root}} = 1$.

Stop when $R_u \le \varepsilon$. Rectifiability gives finitely many first-exit pieces at each node, and (4.2) gives geometric decrease of chord lengths. Thus the tree is finite. Every chain of internal nodes has length at most

$$M_N = 1 + \left\lceil \frac{N}{n-1} \right\rceil . \tag{4.3}$$

For an internal node, define

$$S_u = \sum_{v \text{ child of } u} R_v, \qquad a_u = \log(S_u/R_u), \qquad p_{uv} = R_v/S_u \,. \tag{4.4}$$

Let $I_u = [x_u, y_u]$, $H_u = \langle H_N, \nu_{I_u} \rangle$, and let Q_u be the ordered polygon consisting of the child chords. Then

$$\Delta_u = \sum_v p_{uv} H_v - H_u = \langle H_N, \nu_{Q_u} - \nu_{I_u} \rangle \,. \tag{4.5}$$

The weights $\theta(\text{root}) = 1$ and $\theta(v) = \theta(u)p_{uv}$ define a probability distribution on leaves.

Lemma 4.1. *For every admissible rectifiable path,*

$$\log \int_P e^{\zeta H_N} |dz| \ge \zeta H_{\text{root}} + \sum_{u \text{ internal}} \theta(u)(a_u + \zeta \Delta_u) - \zeta \omega_N \,, \tag{4.6}$$

where

$$\omega_N = \sup_{\substack{z,w \in \overline{U} \\ |z-w| \le 8\varepsilon}} |H_N(z) - H_N(w)| \,.$$

Moreover, $H_{\text{root}} = O_{\mathbb{P}}(1)$ *and* $\omega_N = O_{\mathbb{P}}(\sqrt{N})$.

Proof. Each point of a leaf is within 5ε of every point of its chord, by (4.2). Hence the leaf has cost at least $R_v e^{\zeta H_v - \zeta \omega_N}$. The definitions give

$$R_v = \theta(v) \exp\left(\sum_{u \prec v} a_u \right), \qquad \sum_{v \text{ leaf}} \theta(v) H_v = H_{\text{root}} + \sum_{u \text{ internal}} \theta(u) \Delta_u \,.$$

Sum the leaf costs and apply Jensen's inequality with the leaf probabilities $\theta(v)$. This proves (4.6).

The growth bound for the unit segment gives $\operatorname{Var} H_{\text{root}} \le C_d$ by the calculation in Lemma 3.4(a). For the oscillation, the kernel derivative bound gives

$$\mathbb{E}(H_N(z) - H_N(w))^2 \le C_d |z - w|^2 / \varepsilon^2 \,.$$

The Gaussian family of differences with $|z - w| \le 8\varepsilon$ therefore has bounded variance. At canonical radius $s \le 1$, discretizing z, w with Euclidean mesh $c_d \varepsilon s$ gives a covering number at most $C_d \varepsilon^{-2d} s^{-2d}$. The entropy integral gives $\mathbb{E}\omega_N \le C_d \sqrt{N+1}$, and concentration gives the asserted bound in probability. □

Fix a sufficiently small absolute constant $c_0 > 0$, and put $a_0 = c_0/K$. Call a node good if $a_u \le a_0$ and bad otherwise. The next deterministic estimates are uniform in K.

Lemma 4.2. *A good node has exactly K children. Every edge of Q_u makes angle at most $1/10$ with its parent chord. Both ν_{Q_u} and ν_{I_u} satisfy (3.4) at scale R_u with a universal constant, and they have a displacement coupling of size at most $CR_u\sqrt{a_u}$. If z_i is the ith vertex, then*

$$\left| z_i - \left(x_u + \frac{i}{K}(y_u - x_u) \right) \right| \le CR_u\sqrt{a_u}, \qquad 0 \le i \le K\,. \tag{4.7}$$

Proof. Write ℓ_i for the edge lengths, t_i for their unit directions, and $e = (y_u - x_u)/R_u$. Since $\sum_i \ell_i t_i = R_u e$,

$$\sum_i \ell_i |t_i - e|^2 = 2(S_u - R_u)\,. \tag{4.8}$$

The lower bound $\ell_i \ge R_u/(2K)$ implies $|t_i - e|^2 \le 4K(e^{a_u} - 1)$. Taking c_0 small makes every angle at most $1/10$. It also gives $S_u \in [Kr, Kr + r/4)$.

If there are j children and the final piece was merged, then $S_u \in [(j-1/2)r, (j+1/2)r]$, so $j = K$. If a residual was retained, then $S_u \in ((j-1/2)r, jr)$, which is incompatible with $S_u \in [Kr, Kr+r/4)$. If there is no residual, then $S_u = jr$ and again $j = K$. Thus the first $K-1$ edges have length r. Applying Cauchy–Schwarz to their directional deviations and using (4.8) gives (4.7); the last vertex is the parent endpoint.

The forward angle bound makes the projection on the parent line strictly increasing. The arclength inside a ball of radius t is at most $2t/\cos(1/10)$, which proves the growth bound. Finally parametrize Q_u by normalized arclength on $[0,1]$. Its derivative has constant magnitude S_u and mean $y_u - x_u$, so

$$\int_0^1 |Q_u'(s) - (y_u - x_u)|^2\, ds = S_u^2 - R_u^2\,.$$

Couple equal values of s with the parent segment and use Cauchy–Schwarz. The maximal displacement is at most $\sqrt{S_u^2 - R_u^2} \le CR_u\sqrt{a_u}$. □

4.2 The local Gaussian optimization

We next compare the mean gain of one good polygon with $A_{n,d}$. The parent endpoints will eventually be recorded only to accuracy δ times the chord scale. Their resulting uncertainty must have a cost smaller than n, because $A_{n,d}$ itself is of order at most n.

Fix two cubes of side δR whose centers are at distance in $[R/4, 4R]$. Let the parent endpoints range in these cubes, and allow the good child polygons from the preceding first-exit construction. In particular, each polygon has exactly K edges, the first $K-1$ having length $\mathrm{len}(I)/K$, and its last edge joins the penultimate vertex to the parent endpoint. We retain all the geometric conclusions of Lemma 4.2. For a polygon Q and parent chord I, write

$$Y = -\delta^{-1/2}\langle H_N, \nu_Q - \nu_I \rangle, \qquad E_\delta = \delta^{-2}\log\bigl(\mathrm{len}(Q)/\mathrm{len}(I)\bigr)\,. \tag{4.9}$$

Lemma 4.3. *For $0 < \tau, u < 1$, uniformly in N, the scale and location of the cubes, and their orientation,*

$$\limsup_{\delta \downarrow 0} \mathbb{E}\sup\{Y - (1-\tau)E_\delta\} \le (1+u)^{2/3}(1-\tau)^{-1/3}A_{n,d} + C_d u^{-1/2}\,. \tag{4.10}$$

The constant C_d is independent of n, τ, u. More precisely, for every $\eta > 0$ there is $\delta_0 = \delta_0(n, d, \tau, u, \eta) > 0$ such that the expected supremum is at most the right side of (4.10) plus η whenever $0 < \delta < \delta_0$, for every cutoff and every allowed pair of endpoint cubes.

Proof. First replace H_N by the all-scale field h. This increases the expected supremum by (3.3) and (2.3), with the energy penalty as deterministic mean. This step is uniform in every cutoff and parent scale.

Step 1: fixed endpoints. Normalize the parent chord to $I = [0, e_1]$. Write the transverse vertices as δf_i, $f_0 = f_K = 0$, and identify (f_i) with $f \in V_{n,d}$. The first $K-1$ positive longitudinal increments are

$$\sqrt{K^{-2} - \delta^2 |f_i - f_{i-1}|^2}\,;$$

the last edge joins the penultimate vertex to e_1. This defines a polygon $Q_\delta(f)$ on every fixed bounded subset of $V_{n,d}$ for sufficiently small δ. Expanding the first $K-1$ square roots and the length of the last edge gives, uniformly on compact sets,

$$E_\delta(Q_\delta(f)) \longrightarrow \frac{K}{2} \sum_{i=1}^{K} |f_i - f_{i-1}|^2 = \mathcal{E}(f)\,. \tag{4.11}$$

These polygons belong to the intrinsic good family, uniformly on each fixed compact set for small δ. Indeed, their angles and log excess tend uniformly to zero. Apply the first-exit rule to $Q_\delta(f)$ itself. The first $K-1$ edges are completed first-exit pieces of length $r = 1/K$. The last edge has length $r + O(\delta^2)$ and at least r, since its horizontal increment is at least r. Its residual after one completed piece has length at most $r/2$ and is merged back; if the edge has length exactly r, there is no residual. Thus the rule returns precisely $Q_\delta(f)$.

The horizontal knots differ from i/K by $O_{n,d}(\delta^2)$ on such sets. Their rescaled shifts $b_\delta(f)$ are continuous in f and uniformly bounded on every fixed compact set. Lemmas 3.2 and 3.3 therefore give convergence to $\mathcal{G}_d(f)$ of

$$Z_\delta(Q) = -\delta^{-1/2} \langle h, \nu_Q - \nu_I \rangle$$

as a continuous Gaussian process on each compact set.

We check that the energy penalty permits passage to the full family. Lemmas 4.2 and 3.1 give

$$\sup_{E_\delta \le R} \operatorname{Var} Z_\delta \le C_d \sqrt{R}, \qquad \mathbb{E} \sup_{E_\delta \le R} Z_\delta \le C_{n,d} R^{1/4}\,. \tag{4.12}$$

For the second estimate, (4.7) places the transverse parameters in a ball of radius $C\sqrt{R}$. Each first-edge increment lies in the convex ball $|\delta(f_i - f_{i-1})| \le K^{-1} \sin(1/10)$. The segment between two feasible parameter arrays stays in these balls and in the same transverse parameter ball. Along that segment the square-root denominators are at least $K^{-1}\cos(1/10)$. Differentiating and integrating the parametrization therefore gives a physical vertex displacement at most $C_{n,d}\delta$ times the parameter displacement, using $\delta\sqrt{R \wedge (a_0/\delta^2)} \le \sqrt{a_0}$. The good family has no energies above a_0/δ^2, so this truncation does not change the index set. Changing vertices by at most η changes normalized-arclength parametrizations by at most $C_n\eta$. This follows by matching the ordered vertex times, whose changes are bounded by the changes of edge lengths, and then using their bounded constant speeds. Transport therefore bounds the canonical metric by $C_{n,d}$ times the square root of parameter distance. Volumetric covering and the entropy integral prove the second part of (4.12).

For any fixed positive penalty b, on the shell $k \le E_\delta < k+1$ the mean is $O_{n,d}(k^{1/4})$ and the concentration variance parameter is $O_d(\sqrt{k})$. Integrating the Gaussian tail gives a summable bound $C_{n,d,b} \exp(-c_{n,d,b} k^{3/2})$ for the expected positive part after subtracting bk. These estimates are uniform in small δ. Compact convergence and (2.5) now imply, for $a, b > 0$,

$$\lim_{\delta \downarrow 0} \mathbb{E} \sup_{Q} \{a Z_\delta(Q) - b E_\delta(Q)\} = a^{4/3} b^{-1/3} A_{n,d}\,. \tag{4.13}$$

Here the supremum is over the intrinsic good-polygon family. It contains $Q_\delta(f)$ for every fixed f when δ is sufficiently small.

Step 2: varying endpoints. After normalizing a representative chord, the actual endpoints can be written $x_b = \delta b_0$, $y_b = e_1 + \delta b_1$, where $b = (b_0, b_1)$ ranges in a fixed box of $\mathbb{R}^{2d}$ independent of n. Choose similarities T_b carrying $0, e_1$ to x_b, y_b. For sufficiently small δ, they can be chosen smoothly on the entire fixed endpoint box with

$$|T_b(z) - T_{\widetilde{b}}(z)| \le C_d \delta |b - \widetilde{b}|, \qquad |z| \le 2\,. \tag{4.14}$$

For example, use translation by x_b, dilation by $|y_b - x_b|$, and the minimal rotation from e_1 to $(y_b - x_b)/|y_b - x_b|$.

Set $X_\delta(b, Q) = -\delta^{-1/2}\langle h, (T_b)_*(\nu_Q - \nu_I)\rangle$. Exact similarity invariance on mass-zero measures implies

$$\mathrm{Var}(X_\delta(b, Q) - X_\delta(b, \widetilde{Q})) = \mathrm{Var}(Z_\delta(Q) - Z_\delta(\widetilde{Q}))\,.$$

For fixed Q, couple corresponding points under the two similarities. The universal growth bound for good polygons, (4.14), and transport applied separately to the polygons and chords give

$$\mathrm{Var}(X_\delta(b, Q) - X_\delta(\widetilde{b}, Q)) \le C_d |b - \widetilde{b}|\,.$$

The constant here is independent of n.

Split an increment of X_δ by changing Q first and b second. The inequality $|v + w|^2 \le (1 + u)|v|^2 + (1 + u^{-1})|w|^2$ bounds its canonical metric by that of

$$\sqrt{1 + u}\, Z_\delta(Q) + C_d \sqrt{1 + u^{-1}}\, B(b)\,,$$

where B is an independent sum of $2d$ two-sided Brownian motions in the endpoint coordinates. Its expected supremum over the fixed box is bounded by a dimensional constant. The log excess is unchanged by T_b. Apply (2.3) with mean $-(1 - \tau)E_\delta(Q)$, then (4.13). This yields (4.10). □

4.3 Records of chains

For each node choose $L_u = 2^{-j_u}$ with $R_u \in (L_u/2, L_u]$, and use the lattice of half-open cubes of side δL_u. By (4.2), consecutive scale indices differ by one of $n - 1, n, n + 1$.

For a good node, record its parent endpoint cubes, the bin $k = \lfloor a_u/\delta^2 \rfloor$, the index of the followed child, the next scale, and the cubes of that child's endpoints in the next lattice. The other vertices remain free and are included in the local supremum. At a bad node, choose a child v maximizing $H_u - H_v$ and record its endpoint cubes in the parent lattice. Also record the next scale and the followed child's endpoint cubes in that lattice's finer version. The adverse child and the followed child need not be the same, and no occurrence index is recorded.

Put $G_u = -\Delta_u$ at a good node and $G_u = H_u - H_v$ at a bad node, with v chosen as above. Define

$$Y_u = \delta^{-1/2} G_u, \qquad b_u = \begin{cases} a_u/\delta^2, & u \text{ good}, \\ a_0/\delta^2, & u \text{ bad}. \end{cases} \tag{4.15}$$

Since $-\Delta_u \le \max_v (H_u - H_v)$, we have in both cases

$$a_u + \zeta \Delta_u \ge -\delta^2 (Y_u - b_u)\,. \tag{4.16}$$

Lemma 4.4. *Given the parent scale and endpoint cubes, the number of possible next records for a good node in bin k is at most*

$$e^{C_d n}(k+1)^d\,. \tag{4.17}$$

For a bad node it is at most $C_{n,d}\delta^{-4d}$. In a fixed bad record, the expected supremum of Y_u is at most $C_{n,d}$.

Fix any chain record. Allow all geometric choices in each constituent record, without requiring choices in different nodes to agree. For this enlarged Gaussian family,

$$\sup \operatorname{Var}\left(\sum_i Y_i\right) \le C_d \sum_{i\ good} \sqrt{k_i+1} + \frac{C_{n,d}}{\delta}\, m_{\mathrm{b}}\,, \tag{4.18}$$

where m_{b} is the number of bad nodes. These bounds are uniform in N.

Proof. For a good node, (4.7) places the ith vertex within distance $C\delta L_u\sqrt{k+1}$ of the point at fraction i/K along a representative parent chord. The uncertainty in the parent endpoints has the same bound. A ball of that radius meets at most $C_d K^d (k+1)^{d/2}$ cubes in the next lattice. There are K choices of followed child and three choices of scale. Recording two endpoints gives at most $C_d K^{2d+1}(k+1)^d$ possibilities, which proves (4.17).

At a bad node, all recorded child endpoints lie within a fixed dimensional multiple of the parent ball. The two adverse endpoints in the parent lattice have at most $C_d\delta^{-2d}$ possibilities; the followed endpoints in the next lattice have at most $C_{n,d}\delta^{-2d}$ possibilities. This proves the bad-record count without bounding the number of children.

For good nodes i,ℓ, transport and the displacement bound in Lemma 4.2 give

$$|\operatorname{Cov}(Y_i,Y_\ell)| \le C_d 2^{-|j_i-j_\ell|/2}(k_i+1)^{1/4}(k_\ell+1)^{1/4}\,. \tag{4.19}$$

For a bad node, both measures in the chord difference have the growth bound at parent scale with constant C_n, and their displacement is $O_d(R_u)$. Thus (4.19) remains valid if each factor belonging to a bad node is replaced by $C_{n,d}\delta^{-1/2}$. Choose this constant large enough to cover mixed pairs and pairs of bad nodes, leaving the good-good coefficient unchanged. Since the scale indices strictly increase, the matrix $2^{-|j_i-j_\ell|/2}$ has bounded row sums independent of n. Apply $2ab \le a^2+b^2$ to obtain (4.18).

Finally, fix a deterministic reference chord pair in a nonempty bad record. Its field difference is centered. Moving its four endpoints within their recorded cubes changes the chord measures by displacement at most $C_{n,d}\delta R_u$. After division by $\sqrt{\delta}$, transport gives a canonical metric bounded by $C_{n,d}$ times the square root of the change in the rescaled endpoint coordinates. There are $4d$ bounded coordinates, so their entropy integral is bounded by $C_{n,d}$. Subtracting the reference centered Gaussian therefore proves the asserted expected supremum bound. □

The record families are deterministic. In particular, the bad-node family contains every chord pair with the recorded geometric constraints; it is not conditioned on the adverse child maximizing a random quantity. The enlargement in Lemma 4.4 allows separate geometric choices in different nodes. It makes no assertion of probabilistic independence. Smoothness of H_N and finite-dimensional separable approximation make these suprema measurable. Thus the resulting family includes every field-dependent path and every one of its chains.

4.4 Summing the records

Fix $0 < \tau, u < 1$ and $t > 0$. By Lemma 4.3, for all sufficiently small δ the local penalized good-node mean is at most

$$b_n^* = (1+u)^{2/3}(1-\tau)^{-1/3}A_{n,d} + C_d u^{-1/2} + 1\,. \tag{4.20}$$

In bin k, the unpenalized expected supremum of Y_u is consequently at most $b_n^* + (1-\tau)(k+1)$. For a fixed chain record the geometric index set is a product, so the mean of the supremum of $\sum_i Y_i$ is bounded by the sum of these local bounds and the bad-node bounds. This uses linearity of expectation, not independence.

Apply (2.4) with variance bound (4.18). At a good node subtract the deterministic cost k, which is at most the actual b_u throughout its bin. At a bad node subtract a_0/δ^2. After summing the possible next records, the contributions of a good and a bad node are bounded respectively by

$$Z_{\mathrm{g}} = e^{C_d n + t(b_n^*+1)} \sum_{k\geq 0} (k+1)^d \exp\{-t\tau k + C_d t^2 \sqrt{k+1}\}, \tag{4.21}$$

$$Z_{\mathrm{b}} = C_{n,d}\delta^{-4d} \exp\{tC_{n,d} + C_{n,d}t^2/\delta - ta_0/\delta^2\}\,. \tag{4.22}$$

For fixed n, t, τ, u, the negative term of order δ^{-2} in (4.22) dominates the positive term of order δ^{-1} and the counting factor. Hence $Z_{\mathrm{b}} \to 0$ as $\delta \downarrow 0$.

For a deterministic record $\mathfrak{r}$, let $T_{\mathfrak{r}}$ be its enlarged Gaussian supremum and let $B_{\mathfrak{r}} = \sum_{i \text{ good}} k_i + m_{\mathrm{b}} a_0/\delta^2$. Define the exceptional event $\mathcal{A}(T_0)$ as the countable union of $\{T_{\mathfrak{r}} - B_{\mathfrak{r}} > T_0\}$ over all records. This event is measurable and contains every actual chain with score exceeding T_0, since its actual costs are at least the recorded ones. Markov's inequality and the union bound give

$$\mathbb{P}[\mathcal{A}(T_0)] \leq e^{-tT_0} \sum_{j=1}^{M_N} (Z_{\mathrm{g}} + Z_{\mathrm{b}})^j\,. \tag{4.23}$$

There is only one pair of root endpoint cubes. Empty record families may simply be omitted. Choose

$$T_0 = \frac{M_N}{t}\{\log(1 + Z_{\mathrm{g}} + Z_{\mathrm{b}}) + 2\}\,.$$

Since $\sum_{j=1}^{M} Z^j \leq (1+Z)^M$ for $Z \geq 0$, the right side of (4.23) is at most e^{-2M_N}.

On the complementary event, average the bound over the leaf distribution θ of each path. Every internal node appears with weight $\theta(u)$ in this average. Equations (4.6) and (4.16) imply, simultaneously for all paths,

$$\log \int_P e^{\zeta H_N} |dz| \geq -\delta^2 T_0 + \zeta H_{\mathrm{root}} - \zeta \omega_N\,.$$

Take the infimum over paths. For fixed δ, the last two terms are $o_{\mathbb{P}}(N)$ by Lemma 4.1. Let $N \to \infty$ and use (1.3), recalling that $\log \varepsilon = -N \log 2$. Since $M_N/N \to 1/(n-1)$, this gives

$$\frac{\lambda_d(\delta^{3/2})}{\delta^2} \leq \frac{\log(1 + Z_{\mathrm{g}} + Z_{\mathrm{b}}) + 2}{t(n-1)\log 2}\,. \tag{4.24}$$

Take $\tau = u = n^{-1/4}$ and $t = n^{1/4}$. Then $t\tau = 1$, and

$$C_d t^2 \sqrt{k+1} \leq \frac{1}{2}k + C_d' t^4$$

bounds the logarithm of the series in (4.21) by $C_d n$. The logarithm of the record factor has the same bound. Divide by t in (4.24), use (4.20), and let $\delta \downarrow 0$. This proves (4.1). In particular, Lemma 2.1 gives

$$\limsup_{\zeta \downarrow 0} \frac{\lambda_d(\zeta)}{\zeta^{4/3}} \le \kappa_d \,. \tag{4.25}$$

5 The lower comparison

A lower bound for the exponent follows from constructing paths with small expected cost. We choose a favorable polygon using one band of the field, then replace each of its edges using finer bands. To retain the full variational gain through this iteration, we control the expected weighted occupation measure of the finer path. Its horizontal marginal stays close to uniform measure, so the current field band sees almost the same average after all later refinements.

Recall $e_{n,d}$ and $\beta_{n,d} = 3e_{n,d}$ from (2.7). We prove, for all sufficiently large n,

$$\liminf_{\zeta \downarrow 0} \frac{\lambda_d(\zeta)}{\zeta^{4/3}} \ge \frac{\beta_{n,d} - 7}{(n + g_n)\log 2}, \qquad g_n = \lceil 20 \log_2(n+2) \rceil \,. \tag{5.1}$$

Every constant below is chosen before the refinement depth. The order is: fix d, n and the finite family; choose the geometric constants; then make δ sufficiently small. The depth can then be arbitrarily large.

5.1 Polygons with equal edge lengths

Fix $\mathcal{F}_{n,d} = \{f_1, \dots, f_J\}$ from Lemma 2.2, and abbreviate $e = e_{n,d}$ and $g = g_n$. Set

$$K = 2^{n+g}, \quad L = (n+2)^4, \quad E_0 = e + 1, \quad s = 1 + \delta^2 E_0, \quad q = s/K, \quad \rho = \delta L \,. \tag{5.2}$$

The replacement polygon will have K edges of length exactly q, hence total length exactly s. Equal edge lengths ensure that the same finer scale band becomes the same reference band after rescaling each edge.

Let ψ_K be affine on intervals of length $1/K$, with value $1/K$ at odd mesh points and zero at even mesh points. Fix a unit vector $v \in \mathbb{R}^{d-1}$. Since each interval of the original 2^{-n} mesh contains an even number of fine intervals,

$$\int_0^1 f_j'(x) \cdot v \, \psi_K'(x)\, dx = 0, \qquad \mathcal{E}(f_j + a\psi_K v) = \mathcal{E}(f_j) + a^2/2 \,. \tag{5.3}$$

Lemma 5.1. *For all sufficiently small δ, there are numbers $a_{j,\delta} \ge 0$ and polygons $Q_{j,\delta}$ from 0 to e_1 with the following properties. Every polygon has K edges of length q, all directed forward, with edge angles at most $C_{n,d}\delta$. Its transverse vertices are*

$$\delta\big(f_j(i/K) + a_{j,\delta}\psi_K(i/K)v\big), \qquad 0 \le i \le K \,.$$

Writing x_i for the corresponding horizontal coordinates,

$$|x_i - i/K| \le \delta^2 E_0 \,. \tag{5.4}$$

Furthermore,

$$a_{j,\delta} \longrightarrow a_j := \sqrt{2(E_0 - \mathcal{E}(f_j))}, \qquad \sup_{z \in Q_{j,\delta}} |z_\perp| \le M_0\delta, \quad M_0 \le C_d\sqrt{e+1} \,. \tag{5.5}$$

Proof. For a proposed amplitude a, prescribe the positive horizontal increments by

$$\ell_i(a,\delta) = \sqrt{q^2 - \delta^2\,|(f_j + a\psi_K v)(i/K) - (f_j + a\psi_K v)((i-1)/K)|^2}\,.$$

For bounded a, these are defined for sufficiently small δ. Let $S_\delta(a) = \sum_i \ell_i(a,\delta)$. Expanding the square roots and using (5.3) gives, uniformly on bounded intervals of amplitudes,

$$\frac{S_\delta(a) - 1}{\delta^2} \longrightarrow E_0 - \mathcal{E}(f_j) - a^2/2\,.$$

The limiting expression has a positive zero $a_j \geq \sqrt{2}$ and changes sign there. The intermediate value theorem gives a zero $a_{j,\delta} \to a_j$ of $S_\delta(a) - 1$. Starting from 0, use the increments $\ell_i(a_{j,\delta},\delta)$ and the prescribed transverse increments. Their sum is e_1, and each resulting edge has length q.

Each horizontal increment is at most q. The total deficit from Kq is $s - 1 = \delta^2 E_0$, so every partial sum satisfies (5.4). For fixed n, all transverse increments are $O_{n,d}(\delta/K)$ and $q \geq 1/K$, which gives the angle bound. Finally $\|f_j\|_\infty \leq C\sqrt{e}$ by Poincaré, $a_j \leq C\sqrt{e+1}$, and $\|\psi_K\|_\infty = 1/K$. Increasing M_0 slightly covers all small δ and proves (5.5). □

Write $\nu_j = \nu_{Q_{j,\delta}}$ and $\nu_0 = \mathcal{L}$, regarded as a measure on the unit segment. The reference field band is

$$H = H_{\rho q,\rho}\,. \tag{5.6}$$

Choose the least index attaining the maximum in

$$I = \operatorname*{argmax}_{1\leq j\leq J}\{-\delta^{-1/2}\langle H, \nu_j - \nu_0\rangle\}, \qquad w_{n,d,\delta} = \mathbb{E}\max_{1\leq j\leq J}\{-\delta^{-1/2}\langle H, \nu_j - \nu_0\rangle\}\,. \tag{5.7}$$

The selector is measurable and depends only on the current scale band.

Lemma 5.2. *For every sufficiently large n,*

$$\liminf_{\delta\downarrow 0} w_{n,d,\delta} \geq 4e_{n,d} - 4\,. \tag{5.8}$$

Proof. We compare with the finite Gaussian family in (2.9). Put $g_j = f_j + a_j\psi_K v \in V_{n+g,d}$, so $\mathcal{E}(g_j) = E_0$. By (2.2),

$$\operatorname{Var}(\mathcal{G}_d(g_j) - \mathcal{G}_d(f_j)) = 2\pi a_j \int_0^1 \psi_K(x)\,dx \leq C_d\sqrt{e+1}/K\,.$$

The Gaussian maximum bound therefore makes the loss from padding at most

$$C_d(e+1)^{1/4}\sqrt{\frac{\log(2J)}{K}}\,. \tag{5.9}$$

For fixed n, Lemmas 3.2 and 3.3 apply to the actual equal-edge polygons, since $a_{j,\delta} \to a_j$ and the horizontal knots move by $O_{n,d}(\delta^2)$. In Lemma 3.3, take a compact family containing the finitely many amplitude intervals and a fixed box of rescaled knot shifts. Thus their all-scale expected maximum converges to $\mathbb{E}\max_j \mathcal{G}_d(g_j)$.

Remove the scales below ρq. The polygons have a uniform linear growth bound for small δ, by their forward angle bound. Lemma 3.4(a) and the finite Gaussian maximum estimate bound this loss by

$$C_d\sqrt{\frac{\rho q}{\delta}\log(2J)} = C_d\sqrt{Lq\log(2J)}\,.$$

Remove also the scales above ρ. On the fixed finite family, arclength and moving-knot replacement errors tend to zero even in the all-scale metric, hence also in the coarse-band metric. Lemma 3.4(b), applied to $\mathcal{E}(g_j) = E_0$, bounds the remaining coarse loss by $C_d\sqrt{E_0/L}$.

Consequently,

$$\liminf_{\delta\downarrow 0} w_{n,d,\delta} \geq 4e - 1 - C_d(e+1)^{1/4}\sqrt{\frac{\log(2J)}{K}} - C_d\sqrt{\frac{L\log(2J)}{K}} - C_d\sqrt{\frac{E_0}{L}}\,. \tag{5.10}$$

Here $e \leq C_d n$ and $\log(2J) \leq C_d n 2^n$. The padding $2^g \geq (n+2)^{20}$ beats the factors in the first two errors, while $L = (n+2)^4$ makes the last error tend to zero. For sufficiently large n, each of the three errors is at most one, proving (5.8). □

5.2 An invariant class of occupation measures

For each edge i of $Q_{j,\delta}$, let T_i^j be the similarity carrying the unit segment to that oriented edge, with dilation q and the minimal rotation from e_1 to the edge direction. In particular, for $y \perp e_1$ the horizontal component of the rotated vector has magnitude at most $C_{n,d}\delta|y|$. For a deterministic probability measure μ, put

$$\nu_{j,\mu} = \frac{1}{K}\sum_{i=1}^{K}(T_i^j)_*\mu\,. \tag{5.11}$$

The equal edge lengths give $\nu_{j,\nu_0} = \nu_j$.

Write W_∞ for the infimum over couplings of the essential supremum of the displacement. For measures on the line this is the essential supremum of the difference of their quantile functions. Given constants $M, B > 0$, define

$$V_\delta = [-B\delta^2, 1 + B\delta^2] \times \{y \in \mathbb{R}^{d-1} : |y| \leq M\delta\} \tag{5.12}$$

and let $\mathcal{C}_\delta$ be the class of deterministic probability measures supported in V_δ whose horizontal marginal μ_x satisfies

$$W_\infty(\mu_x, \mathcal{L}) \leq B\delta^2\,. \tag{5.13}$$

These measures may be singular or have atoms.

The expected weighted replacement defines the deterministic transformation

$$\mathcal{T}\mu(A) = \frac{\mathbb{E}\int_A e^{\delta^{3/2}H(z)}\,\nu_{I,\mu}(dz)}{\mathbb{E}\int e^{\delta^{3/2}H(z)}\,\nu_{I,\mu}(dz)}\,, \tag{5.14}$$

where I is the selector in (5.7). We require the same class to work for every subsequent refinement, which is why the estimates in the following lemma are uniform over μ.

Lemma 5.3. *For every sufficiently large n, one can choose $M \leq C_d\sqrt{n+1}$ and $B < \infty$, then $\delta_0 > 0$, so that for $0 < \delta < \delta_0$:*

(i) Every edge similarity satisfies $T_i^j(V_\delta) \subset V_\delta$, and $V_\delta \subset U$.

(ii) $\mathcal{T}\mu \in \mathcal{C}_\delta$ for every $\mu \in \mathcal{C}_\delta$.

(iii) Uniformly for $\mu \in \mathcal{C}_\delta$,

$$\mathbb{E} \int e^{\delta^{3/2} H(z)} \, \nu_{I,\mu}(dz) \le 1 - \delta^2(4e_{n,d} - 5) + o_{n,d}(\delta^2) \,. \tag{5.15}$$

The constants and the remainder are independent of the eventual refinement depth.

Proof. Write $q_0 = 2/K$, so $q \le q_0$ for sufficiently small δ. Choose M so that

$$M(1 - q_0) > M_0 + q_0, \qquad M \le C_d \sqrt{n+1} \,. \tag{5.16}$$

For sufficiently large n, we also have $L \ge 2(M+1)$. We will choose B after estimating the change of horizontal marginal. All intermediate estimates below are made under $B\delta \le 1$, $M\delta \le 1$, $\rho \le 1$, and sufficiently small δ depending on n, d.

Step 1: the unweighted replacement. Couple μ_x to $\mathcal{L}$ with horizontal displacement at most $B\delta^2$, then disintegrate μ over its horizontal marginal. This lifts the coupling to μ and ν_0, with total displacement at most $M\delta + B\delta^2$. The existence of this disintegration only uses that the supports are compact subsets of Euclidean space. Applying each T_i^j gives a coupling of $\nu_{j,\mu}$ and ν_j with displacement

$$u = q\delta(M + B\delta) \le q\delta(M+1) \le \rho q/2 \,. \tag{5.17}$$

Thus Lemma 3.4(c),(d) applies at the lower band cutoff ρq, with constants independent of B.

For the horizontal coordinate, the small edge angle gives the stronger bound

$$W_\infty((\nu_{j,\mu})_x, \mathcal{L}) \le \delta^2 \{E_0 + q(B + C_{n,d}M)\} \,. \tag{5.18}$$

Indeed, relative to a point $(u, 0)$ on the reference segment, the horizontal part of $qO_i(x-u, y)$ is at most $qB\delta^2 + qC_{n,d}M\delta^2$. The horizontal marginal of ν_j is within $\delta^2 E_0$ of $\mathcal{L}$, since normalized arclength visits the equal-length edges with mass $1/K$ and (5.4) holds at every knot. This proves (5.18). It also holds for the deterministic mixture $\bar{\nu} = \mathbb{E}\nu_{I,\mu}$.

Step 2: the effect of exponential weighting. All measures involved lie in a fixed compact box for the stipulated small δ. Let Z_δ be the supremum of $|H|$ on this box. The point variance is $\log(1/q) \le \log K$, and the derivative variance is at most $C_d(\rho q)^{-2}$. The entropy integral and Gaussian concentration give

$$\mathbb{E}Z_\delta \le C_{n,d}\sqrt{\log(1/\delta)}, \qquad \mathbb{E}[Z_\delta^2 e^{\delta^{3/2} Z_\delta}] = O_{n,d}(\log(1/\delta)) \,. \tag{5.19}$$

For the second assertion, use Cauchy–Schwarz, the fourth-moment bound from concentration, and $\mathbb{E}e^{tZ_\delta} \le \exp(t\mathbb{E}Z_\delta + C_{n,d}t^2)$. The remainder in the exponential Taylor expansion is therefore $o_{n,d}(\delta^2)$, uniformly over all these probability measures, including the selected one.

Apply (3.15) to the J deterministic measures $\nu_{j,\mu}$, using (5.17). Multiplying its bound by $\delta^{3/2}$ gives

$$\delta^{3/2}\mathbb{E} \sup_{j,x} \left| \int_{\{z_1 \le x\}} H(z) \, \nu_{j,\mu}(dz) \right| \le C_d \delta^2 \sqrt{L}(1 + \sqrt{\log(2J)}) \,.$$

Together with (5.19), this bounds, uniformly in x, the difference between the expected weighted prefix and $\bar{\nu}\{z_1 \le x\}$ by $C'_{n,d}\delta^2 + o_{n,d}(\delta^2)$. The same holds for the total mass. Normalization in (5.14) consequently gives

$$\sup_x |F_{(\mathcal{T}\mu)_x}(x) - F_{\bar{\nu}_x}(x)| \le C'_{n,d}\delta^2 \tag{5.20}$$

for sufficiently small δ, after increasing $C'_{n,d}$. This constant is independent of B: the prefix estimate uses only (5.17), and (5.19) uses a fixed box.

We use the following elementary quantile observation. If $W_\infty(\sigma, \mathcal{L}) \le r$, a probability measure $\widetilde{\sigma}$ has support in $[-r, 1+r]$, and their distribution functions differ by at most η, then

$$W_\infty(\widetilde{\sigma}, \mathcal{L}) \le r + \eta\,.$$

For quantile levels in $[\eta, 1-\eta]$, sandwich the quantile of $\widetilde{\sigma}$ between those of σ at levels shifted by η. The support assumption handles the remaining levels. This proof also covers atoms. In our application exponential weighting preserves the support bound of $\bar{\nu}_x$. Apply the observation to (5.18) and (5.20). We obtain

$$W_\infty((\mathcal{T}\mu)_x, \mathcal{L}) \le \delta^2\{E_0 + q_0 B + q_0 C_{n,d} M + C'_{n,d}\}\,. \tag{5.21}$$

Choose

$$B > \frac{E_0 + q_0 C_{n,d} M + C'_{n,d}}{1 - q_0}\,. \tag{5.22}$$

After this choice, decrease δ_0 to impose $B\delta \le 1$ and all the small δ conditions used above. This makes (5.21) at most $B\delta^2$.

Step 3: deterministic containment. Take $(x, y) \in V_\delta$ and clamp x to $u \in [0, 1]$. The point $T_i^j(u, 0)$ lies on the corresponding edge, whose horizontal coordinate belongs to $[0, 1]$. The horizontal displacement from this point is at most $q\delta^2(B + C_{n,d}M)$. It is at most $B\delta^2$ by (5.22). The transverse coordinate of $T_i^j(x, y)$ is bounded by

$$\delta(M_0 + qM + qB\delta) \le \delta(M_0 + q_0 M + q_0) < M\delta$$

by (5.16). Hence every T_i^j maps V_δ into itself. Also $V_\delta \subset U$ when δ is sufficiently small. This proves (i), supplies the support part of (ii), and, together with (5.21), proves (ii).

Step 4: the gain after refinement. Lemma 3.4(c) and (5.17) give

$$\operatorname{Var}\langle H, \nu_{j,\mu} - \nu_j\rangle \le C_d \frac{q\delta(M + B\delta)^2}{L}\,.$$

The finite Gaussian maximum estimate yields

$$\limsup_{\delta \downarrow 0} \sup_{\mu \in \mathcal{C}_\delta} \delta^{-1/2} \mathbb{E} \max_j |\langle H, \nu_{j,\mu} - \nu_j\rangle| \le C_d M \sqrt{\frac{\log(2J)}{KL}}\,. \tag{5.23}$$

This tends to zero as $n \to \infty$, since $M \le C_d\sqrt{n+1}$, $\log(2J) \le C_d n 2^n$, and $K = 2^{n+g}$. Increase the threshold for n so that it is at most one.

The definition of the selector gives the exact identity

$$\mathbb{E}\langle H, \nu_I\rangle = -\sqrt{\delta}\, w_{n,d,\delta}\,,$$

because $\langle H, \nu_0\rangle$ is centered. No independence between I and this common field average is needed. Lemma 5.2 and (5.23) therefore imply, uniformly over deterministic $\mu \in \mathcal{C}_\delta$,

$$\mathbb{E}\langle H, \nu_{I,\mu}\rangle \le -\sqrt{\delta}\,(4e - 5) + o_{n,d}(\sqrt{\delta})\,.$$

Apply the exponential expansion once more, with the uniform remainder from (5.19). This proves (5.15). □

5.3 Iteration and the exponent

For $j \geq 0$, let

$$X_j = H_{\rho q^{j+1},\rho q^j}\,. \tag{5.24}$$

These are independent fields. Construct P_r by r rounds of replacements, starting with the unit segment. At round j, each edge has length q^j; use X_j in that edge's coordinates and apply the selector (5.7). Carry the coordinate frames recursively: the frame of a child is $O_{\text{child}} = O_{\text{parent}}O_{\text{local}}$, where O_{local} is the chosen edge rotation. The frame at a node is consequently measurable with respect to the previously used bands. Every possible path lies in V_δ by Lemma 5.3(i).

Define the expected weighted occupation measure

$$M_r(A) = \mathbb{E}\int_{P_r} \mathbf{1}_A(z)\exp\left(\delta^{3/2}\sum_{j=0}^{r-1} X_j(z)\right)|dz|, \qquad m_r = M_r(\mathbb{R}^d), \qquad \mu_r = M_r/m_r\,. \tag{5.25}$$

For $r = 0$, these definitions give $m_0 = 1$ and $\mu_0 = \nu_0$. Every μ_r is deterministic. The total mass is positive and finite: the Euclidean length of P_r is exactly s^r, and the supremum of the smooth field $H_{\rho q^r,\rho}$ on the fixed box has finite exponential moments.

Lemma 5.4. *For all $r \geq 1$,*

$$m_r = s\,m_{r-1}\,\mathbb{E}\int e^{\delta^{3/2}H}\,d\nu_{I,\mu_{r-1}}, \qquad \mu_r = \mathcal{T}\mu_{r-1}\,. \tag{5.26}$$

For sufficiently small δ, with n fixed as above,

$$m_r \leq \exp\{-r\delta^2(\beta_{n,d} - 7)\}, \qquad r \geq 1\,. \tag{5.27}$$

Proof. Condition on the entire first band $X_0 = H$. This fixes the selected polygon and its edge similarities T_i^I. The finer bands are independent of this conditioning. Since every edge has dilation q, their pullbacks by T_i^I have the same joint law as the reference bands used in a depth $r-1$ construction. The recursively carried frames make the selection rule agree exactly in those coordinates. Translation and rotation invariance remove the position and orientation of the selected edge.

The conditional expected finer occupation measure on edge i is therefore $q(T_i^I)_*M_{r-1}$. Multiplication by the remaining coarse factor $e^{\delta^{3/2}H}$ and summation over the edges give

$$M_r(A) = q\,m_{r-1}\mathbb{E}\sum_{i=1}^{K}\int \mathbf{1}_A(T_i^I z)e^{\delta^{3/2}H(T_i^I z)}\,\mu_{r-1}(dz)\,.$$

Using $Kq = s$ proves both identities in (5.26). This calculation uses only conditional expectation for each edge and linearity of the sum. Finer fields on different edges may be correlated.

Since $\mu_0 \in \mathcal{C}_\delta$, Lemma 5.3(ii) and the second identity imply $\mu_r \in \mathcal{C}_\delta$ for all r. Apply its uniform saving estimate in the first identity. With $s = 1 + \delta^2(e+1)$ and $\beta_{n,d} = 3e$,

$$\begin{aligned}\log(m_r/m_{r-1}) &\leq \log(1+\delta^2(e+1)) + \log\bigl(1 - \delta^2(4e-5) + o_{n,d}(\delta^2)\bigr)\\ &\leq -\delta^2(\beta_{n,d} - 6) + o_{n,d}(\delta^2)\,.\end{aligned}$$

The remainder is independent of r. Decrease δ so that it is at most δ^2. Induction from $m_0 = 1$ proves (5.27). □

Set $\varepsilon_r = \rho q^r$. The field in (5.25) is $H_{\varepsilon_r,\rho}$, so m_r is the expected $H_{\varepsilon_r,\rho}$-length of P_r. For any $\eta > 0$, Markov's inequality and (5.27) give

$$\mathbb{P}\left[\log \int_{P_r} e^{\delta^{3/2} H_{\varepsilon_r,\rho}} |dz| > -r\delta^2(\beta_{n,d} - 7) + \eta r\right] \le e^{-\eta r}\,.$$

For fixed n, δ, the omitted field $H_{\rho,1}$ is smooth on $\overline{U}$ and its supremum is $O_{\mathbb{P}}(1)$ independently of r. The cost of P_r for $H_{\varepsilon_r,1}$ is at most its displayed band cost multiplied by $\exp(\delta^{3/2} \sup_U H_{\rho,1})$. All these paths are admissible. Use (1.3) along ε_r and then let $\eta \downarrow 0$. We obtain

$$\frac{\lambda_d(\delta^{3/2})}{\delta^2} \ge \frac{\beta_{n,d} - 7}{\log(1/q)}\,. \tag{5.28}$$

The cutoff is removed here with n, δ fixed. Next let $\delta \downarrow 0$, so $\log(1/q) \to (n + g_n)\log 2$. This proves (5.1). Finally $\beta_{n,d} = A_{n,d} + O_d(\sqrt{n})$ and $g_n = O(\log n)$ yield

$$\liminf_{\zeta \downarrow 0} \frac{\lambda_d(\zeta)}{\zeta^{4/3}} \ge \kappa_d\,. \tag{5.29}$$

A Normalization and the fixed-parameter exponent

We record the fixed-parameter input and its compatibility with the cutoff used in the proof. This also identifies the planar exponent appearing in the positivity input.

A.1 From medians to convergence in probability

The white-noise field in (1.1), at cutoff 2^{-N}, is exactly the field in [2, Section 1.1, equation (1.4)]. The same paper writes $B_2(0) = (-2, 2)^d$ and uses the median of the internal distance from 0 to e_1 in that cube. By [2, Proposition 1.1], for every fixed $\zeta > 0$ there is $Q_d(\zeta)$ such that

$$\operatorname{median} D^{\zeta}_{2^{-N},d} = 2^{-N(1-\zeta Q_d(\zeta)) + o(N)}\,. \tag{A.1}$$

Their path convention is piecewise C^1. It gives the same infimum as (1.2): at a fixed cutoff the weight is smooth and positive, and any rectifiable path can be approximated by polygonal paths in the open cube with convergent weighted lengths. Its compact image has positive distance from the boundary, so the approximation stays in the cube. Rational polygonal approximation also gives measurability of the infimum.

A Cameron–Martin shift g of the underlying noise with norm $\|g\|_2$ changes each field value by at most $\sqrt{N \log 2}\,\|g\|_2$. Comparing all path costs therefore gives

$$|\log D^{\zeta}_{2^{-N},d}(W + g) - \log D^{\zeta}_{2^{-N},d}(W)| \le \zeta\sqrt{N \log 2}\,\|g\|_2\,.$$

Gaussian concentration about a median implies that the difference between $\log D^{\zeta}_{2^{-N},d}$ and the logarithm of its median is $O_{\mathbb{P}}(\sqrt{N})$ for fixed ζ. Combining this with (A.1) proves (1.3) along dyadic cutoffs, with

$$\lambda_d(\zeta) = 1 - \zeta Q_d(\zeta)\,. \tag{A.2}$$

For $\varepsilon \in [2^{-N-1}, 2^{-N}]$, the difference between $H_{\varepsilon,1}$ and $H_{2^{-N},1}$ is a band with point variance at most $\log 2$ and derivative variance at most $C_d 2^{2N}$. Its supremum on $\overline{U}$ is $O_{\mathbb{P}}(\sqrt{N})$ by the same entropy estimate as in Lemma 4.1, uniformly in this range of ε. Comparing path costs by that supremum proves (1.3) for arbitrary cutoffs. In particular it applies to the sequence $\varepsilon_r = \rho q^r$ used in Section 5. Omitting or adding a fixed coarse band changes log distances by $O_{\mathbb{P}}(1)$ and has no effect on the exponent.

A.2 Comparison with mollified fields

In this subsection, constants may also depend on the fixed mollifier, the normalizing probability measure, and the compact set under consideration. Let h be the all-scale logarithmic field of (3.2), normalized by its pairing with a fixed smooth compactly supported probability measure ν_* being zero. Let φ be a smooth compactly supported probability density, and put $\varphi_\varepsilon(z) = \varepsilon^{-d}\varphi(z/\varepsilon)$. For a compact set $V \subset \mathbb{R}^d$, couple $h * \varphi_\varepsilon$ and $H_{\varepsilon,1}$ using the same noise.

Lemma A.1. *For this coupling,*

$$\sup_{z\in V} |(h * \varphi_\varepsilon)(z) - H_{\varepsilon,1}(z)| = O_{\mathbb{P}}(\sqrt{\log \varepsilon^{-1}}) \,. \tag{A.3}$$

In dimension two the same conclusion holds for circle averages of radius ε, including the normalization that the unit-circle average at the origin is zero. Consequently these regularizations have the same fixed-parameter LFPP exponent.

Proof. For a point $x \in V$, split the difference of the fields into the scales below ε, between ε and one, and above one, together with the normalization term. For $t \le \varepsilon$, Young's inequality gives

$$\|K_t * \varphi_\varepsilon\|_2^2 \le C_d t^{d-1}\varepsilon^{-d} = C_d t^{-1}(t/\varepsilon)^d \,.$$

Its integral over $(0, \varepsilon)$ is bounded. For $t \ge \varepsilon$, the L^2 translation bound for K_t gives

$$\|K_t * \varphi_\varepsilon - K_t\|_2^2 \le C_d \varepsilon^2 t^{-3} \,,$$

whose integral over $(\varepsilon, 1)$ is also bounded. Above scale one, the measures $\varphi_\varepsilon(x - \cdot)$ and ν_* have equal mass and uniformly bounded support diameter for $x \in V$. Their kernel difference has squared L^2 norm at most $C_{d,V} t^{-3}$, which is integrable at infinity. The normalization contribution below one has bounded variance since ν_* is smooth. These bounds prove a uniform bound for the point variance of the remainder.

The point variance of its derivative is $O_{d,V}(\varepsilon^{-2})$. For the scales below ε, this follows from $\|K_t * \nabla\varphi_\varepsilon\|_2^2 \le C_d t^{d-1}\varepsilon^{-d-2}$; above ε, use $\|\nabla K_t\|_2^2 \le C_d t^{-3}$. The canonical metric of the remainder is therefore bounded by $C_{d,V} \min(1, |x-y|/\varepsilon)$. Its entropy integral on V is $O_{d,V}(\sqrt{\log \varepsilon^{-1}})$, and Gaussian concentration proves (A.3).

For circle averages in dimension two, let $\sigma_{\varepsilon,x}$ be uniform measure on the radius ε circle about x. Compact support of C and linear growth along a circle give

$$\|K_t * \sigma_{\varepsilon,x}\|_2^2 \le C/\varepsilon, \qquad t \le \varepsilon \,.$$

The middle and coarse-scale estimates above use only translation bounds and bounded support, so they remain valid. Hence the remainder still has bounded point variance. At distances $|x - y| \le \varepsilon$, Lemma 3.1 applied to the two circle measures gives increment variance at most $C|x-y|/\varepsilon$ for the circle-average field. The cutoff field has increment variance at most $C|x-y|^2/\varepsilon^2$. The entropy integral for their difference again gives (A.3). Changing from the smooth normalization ν_* to zero unit-circle average adds a single finite-variance Gaussian constant.

Finally, if two fields differ by at most M_ε on U, every path cost and hence every distance differs by a multiplicative factor between $e^{-\zeta M_\varepsilon}$ and $e^{\zeta M_\varepsilon}$. In (A.3), $M_\varepsilon/\log \varepsilon^{-1} \to 0$ in probability, which proves equality of the exponents. □

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